MILITARY SPECIFICATION

FRAME, RUCKSACK, STEEL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of steel rucksack frame (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces For Organic Coatings.
TT-E-527	-	Enamel, Alkyd, Lustreless.
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

FSC 8465

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DRAWINGS

ARMY NATICK LABORATORIES

2-3-81 - Frame, Rucksack, Steel.

(Miniature copy of Drawing 2-3-81, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 513 - Electric-Resistance Welded Carbon and Alloy Steel Mechanical Tubing.

A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample frame covered by this specification shall be submitted or made available to the contracting officer or his authorized representative

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for approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.- Materials shall conform to the referenced specifications, drawing and requirements specified herein.

3.2.1 Steel.-

3.2.1.1 Tubing.- The steel tubing shall be electric-resistance welded or seamless, steel numbers MT 1015 or MT 1020 conforming to ASTM A 513 or ASTM A 519, as applicable.

3.2.1.2 Sheet or strip.- Steel sheet shall conform to cold rolled, commercial quality, and steel strip shall conform to cold rolled, skin passed temper (No. 4), rounded edges of QQ-S-698.

3.2.2 Primer.- The primer shall conform to TT-P-636.

3.2.3 Enamel.- The enamel shall conform to TT-E-527 and shall be Olive drab color No. X34087 of FED-STD-595.

3.3 Design and construction.- Design and construction of the frame shall be as shown on Drawing 2-3-81 and as specified herein. The body and sleeve shall be fabricated of tubing specified in 3.2.1.1. The reinforcement and cross plates shall be fabricated of steel sheet or strip specified in 3.2.1.2. The frame shall be free from joint failure, cracked weld or permanent deformation greater than 1/4 inch and the crimped-on sleeve shall not slip or become loose when tested as specified in 4.4.1.

3.4 Welding.- All welding shall be sound and free from cracks, pits, flash, burn through area, hole or fissure.

3.5 Finish.-

3.5.1 Pretreatment.- After complete fabrication, the entire frame shall be given a pretreatment conforming to any method, type I or II of TT-C-490.

3.5.2 Priming.- After pretreatment, all surfaces shall be completely coated with primer specified in 3.2.2.

3.5.3 Enameling.- After the primer has been allowed to dry, primed surfaces shall be completely coated with enamel specified in 3.2.3 and baked. The finish coating shall be uniform in appearance, dry, adherent and free of peels, flakes, scratch or scuff, sags, runs, pinholes, blisters, wrinkles, bare spots with metal showing through, foreign matter embedded in finish, wet or tacky paint. The supplier shall touch up areas of finish damaged during examination.

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3.6 Marking for identification.- Prior to finishing, the frame shall be permanently and legibly indented with the letter "U.S." and the supplier's name, trade name or trademark in a suitable location, readily visible. The marking shall be legible after finishing.

3.7 Workmanship.- All components of the finished frame shall be free from split, puncture, fracture, looseness, dent, sharp edge or burr and shall not be bent out of shape, malformed, misplaced or out of alignment. The tubing shall be free from any flattened area out of round more than 20 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in table II and dimensions specified, and tested as specified in 4.4.1. Any visual defect, any dimension not within specified requirements or any failure to meet test requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawing and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process examination.- Examination shall be made during the manufacturing process for the requirements listed in table I to establish that no deviation is made from the indicated requirements. Whenever a nonconformance is noted, correction shall be made to affected items and process.

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TABLE I.- Process examination

Requirement or operation	Requirement paragraph
Welds are sound, free from cracks, pits, flash, burn through area, hole or fissure.	3.4
Pretreatment of frame prior to application of primer.	3.5.1
Application of primer prior to enameling.	3.5.2
Primer is dry prior to enameling.	3.5.3
Marking prior to finishing.	3.6

4.3.3 End item inspection. The lot shall consist of all of the completely fabricated frames offered for inspection at one time. The sample unit shall be one frame.

4.3.3.1 Visual examination.- Visual examination shall be made of the frames for the defects listed in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished.	X	
	Color of finish is not as specified.		X
	Coating is not uniform, i.e., sags, runs, pinholes, blisters or wrinkles.		X
	Coating not adherent, e.g., peeled or flaked.	X	
	Scratched or scuffed exposing bare metal.	X	
	Scratched or scuffed exposing primer.		X
	Bare spots with metal showing through.	X	
	Foreign matter imbedded in finish.		X
	Finish is not dry, i.e., wet or tacky to touch.		X

NOTE: Finish damaged during examination shall not be scored as a defect. However, the supplier shall touch up areas of finish damaged during examination.

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship	Any part or component not fabricated of applicable referenced materials.	X	
	Any component part missing or not fabricated as specified, e.g., sleeve not crimped as shown.	X	
	Any component split, punctured, fractured or has sharp edge.	X	
	Any component dented, bent out of shape or malformed.		X
	Any component misplaced or out of alignment.	X	
	Any component loose or has burrs.		X
	Operation omitted.	X	
	Machine or die mark affecting appearance.		X
	Tubing has flattened area out of round more than 20 percent.		X
Welding	Missing where required.	X	
	Not type specified.	X	
	Not sound e.g., evidence of crack.	X	
	Evidence of pits, flash, burn through area, hole or fissure.		X
Marking	Missing, illegible, incorrect, incomplete or not in suitable location.		X

4.3.3.2 Dimensional examination.- Examination shall be made of the frames for defects in dimensions. Any dimension not within specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.4 End item testing.- The finished frame shall be tested as specified in 4.4.1. The lot shall consist of all of the completely fabricated frames offered for testing at one time. The sample unit shall be one frame. The inspection level shall be S-1 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.3.5 Examination of preparation for delivery.- An examination shall be made to determine that packing and markings are in compliance with section 5 of this specification. Defects shall be scored as specified in table III.

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The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examine	Defect
Marking	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component, component missing, damaged or otherwise defective.
Workmanship	Bulged or distorted container. Frames not arranged as specified e.g., not ten in length in an upright position with every other frame nested alternately reversed top to bottom. Frame not separated from another with paper. Contents of shipping container not completely covered on top and bottom with paper.

4.3.6 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and strapping shall be in accordance with the appendix of the box specification.

4.3.7 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be scored as specified in table IV. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE IV.- Examination of palletization

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not banded with required straps as specified.

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TABLE IV.- Examination of palletization (cont'd)

Examine	Defect
Weight	Exceeds maximum load limit.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.4 Test.-

4.4.1 Frame test.- The frame shall be measured then placed on a flat, level surface and a 350 pound compressive load shall be applied (at a load rate not to exceed 350 pounds per minute) as shown on Drawing 2-3-81. The test block and board shall be any type of wood. The board shall be of sufficient size to completely cover the sleeve and reinforcement end of frame (approximately 18 by 12 inches) and of sufficient thickness to withstand the 350 pound load. After completion of test, the load shall be removed and the frame shall be examined for the requirements specified in 3.3. Any measurement not within 1/4 inch of measurement before testing or any joint failure, cracked weld, slippage or looseness of a crimped-on sleeve shall be classified as a defect.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.- Ten frames shall be packed in a close-fitting fiberboard box conforming to grade V2s, style RSC-L of PPP-B-636. The inside of each box shall be fitted with a taped liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-F-320. The frames shall be arranged ten (10) in length in an upright position with every other frame nested and alternately reversed top to bottom within a shipping container. Each frame shall be separated from the others with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Each group of nested frames shall be secured compactly by crosstying with cotton tape or twine. Each shipping container shall have the contents completely covered on the top and bottom with a sheet of the same kraft paper as specified above. Each fiberboard shipping container shall be closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix of the box specification.

5.1.2 Level B.- Ten frames shall be packed in a close-fitting fiberboard box conforming to type CF (variety SW) or SF, class domestic, grade 275, style RSC-L of PPP-B-636. The inside of each box shall be fitted with a taped liner conforming to type CF, variety SW, class domestic, grade 275 of PPP-F-320. The frames shall be arranged ten (10) in length in an upright position with

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every other frame nested alternately reversed top to bottom within a shipping container. Each frame shall be separated from the others with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Each shipping container shall have the contents completely covered on top and bottom with a sheet of the same kraft paper as specified above. Each container shall be closed in accordance with method II as specified in the appendix of the box specification.

5.1.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification.

5.1.3 Level C.- Frames shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Palletization.- Unless otherwise specified (see 6.2), frames, packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern intended to be used shall be submitted to and approved by the contracting officer prior to its use.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The rucksack frame is intended to be used as a component of a mounting arrangement for radio sets AN/PRC 41, AN/PRC 47 and VRC 46.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When preproduction sample is not required (see 3.1).
(c) Selection of applicable levels of packing (see 5.1).
(d) When weather-resistant fiberboard boxes are required for level B shipment (see 5.1.2.1).
(e) When palletization is not required (see 5.2).

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6.3 Supersession data.- This specification includes the requirements of Natick Laboratories Interim Purchase Description ED/DES S-42-7 dated 13 September 1967.

Custodians:

Army - GL

Navy - MC

Air Force - 82

Review activities:

Navy - MC

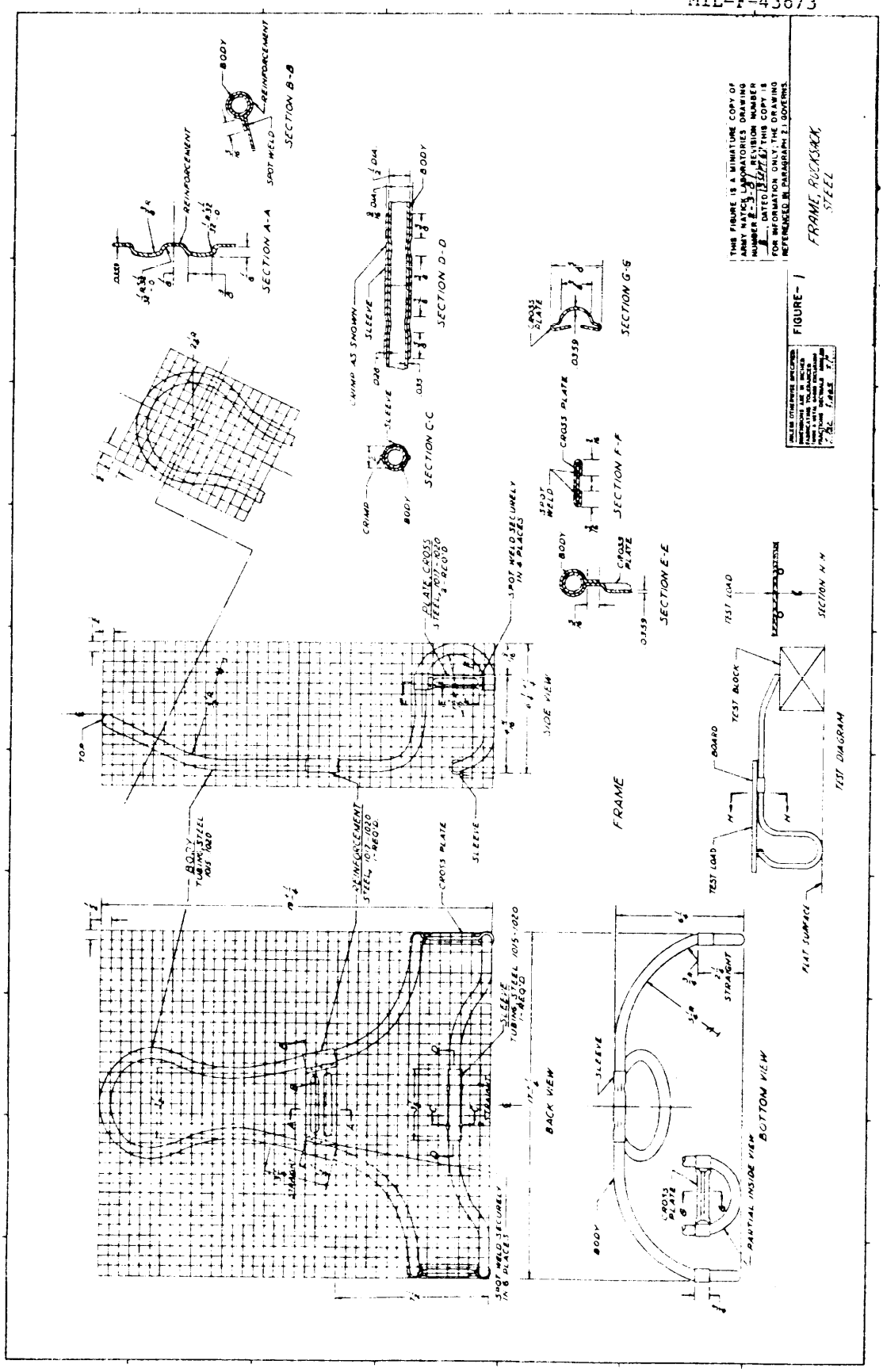
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Preparing activity:

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) - Optional	b. WORK TELEPHONE NUMBER (Include Area Code) - Optional
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